Derivation of the Hamilton-Jacobi-Bellman Equation for the Informal Worker

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Step 1: *Setup.*

Consider an informal worker in continuous time. The agent's state is characterized by assets a and productivity z. The agent faces three independent Poisson processes:

- Job destruction with arrival rate γ^d.
- Productivity change with arrival rate γ^z.
- Formal job offer with arrival rate γ^f.

Additionally, the agent can freely choose to leave informal employment to become unemployed or an entrepreneur (formal or informal) at any time.

Step 2: *Bellman Equation in Discrete Time.*

Consider a period of small length Δ. We define:

$$\beta(\Delta) = e^{-\rho\Delta} \approx 1 - \rho\Delta,$$

where ρ is the subjective discount rate.

Over an interval Δ, since the Poisson processes are independent, the probabilities of each event are:

$$\Pr[\text{job destruction}] = \gamma^d\Delta + o(\Delta), \tag{1}$$
$$\Pr[\text{productivity change}] = \gamma^z\Delta + o(\Delta), \tag{2}$$
$$\Pr[\text{formal job offer}] = \gamma^f\Delta + o(\Delta), \tag{3}$$
$$\Pr[\text{no event}] = 1 - (\gamma^d + \gamma^z + \gamma^f)\Delta + o(\Delta), \tag{4}$$

where $o(\Delta)/\Delta \to 0$ as $\Delta \to 0$. The probability that two or more events occur simultaneously is of order $o(\Delta)$ and therefore vanishes in the limit.

The law of motion for assets is:

$$a_{t+\Delta} = a_t + \Delta \cdot \dot{a}, \qquad \text{with } \dot{a} = w \cdot \upsilon(z) + ra - c + b^i, \tag{5}$$

where w is the informal wage per efficiency unit, $\upsilon(z)$ denotes the agent's effective units of labor, r is the interest rate, c is consumption, and b^i is the social benefit for the informal worker.

The Bellman equation in discrete time is:

$$V^i(a,z) = \max_c \Big\{ u(c)\Delta + \beta(\Delta) \cdot \mathbb{E}\big[V^i(a_{t+\Delta}, z_{t+\Delta})\big] \Big\}, \tag{6}$$

subject to $a_{t+\Delta} \geq \underline{a}$.

The expected value decomposes all possible events:

$$\begin{aligned} \mathbb{E}\big[V^i(a_{t+\Delta}, z_{t+\Delta})\big] = &\big[1-(\gamma^d+\gamma^z+\gamma^f)\Delta\big] \cdot V^i(a_{t+\Delta}, z) \\ &+ \gamma^d\Delta \cdot V^u(a_{t+\Delta}, z) \\ &+ \gamma^z\Delta \cdot \int V^i(a_{t+\Delta}, z')\, d\Pr_z(z'|z) \\ &+ \gamma^f\Delta \cdot \max\big\{V^f(a_{t+\Delta}, z),\ V^i(a_{t+\Delta}, z)\big\}. \end{aligned} \tag{7}$$

The four terms capture, respectively: (i) no event occurs and the agent remains an informal worker with the same productivity; (ii) the job is destroyed and the agent transitions to unemployment; (iii) productivity changes from z to a new value z' drawn from the conditional distribution $\Pr_z(z'|z)$; and (iv) a formal job offer arrives, which the agent accepts only if it is beneficial (hence the max operator).

Step 3: *Substitution and Expansion.*

We substitute (7) into (6) and use the approximation $\beta(\Delta) \approx 1 - \rho\Delta$:

$$\begin{aligned} V^i(a,z) = \max_c \Big\{ u(c)\Delta + (1-\rho\Delta)\Big[&\big(1-(\gamma^d+\gamma^z+\gamma^f)\Delta\big) \cdot V^i(a_{t+\Delta}, z) \\ &+ \gamma^d\Delta \cdot V^u(a_{t+\Delta}, z) + \gamma^z\Delta \int V^i(a_{t+\Delta}, z')\, d\Pr_z(z'|z) \\ &+ \gamma^f\Delta \cdot \max\big\{V^f(a_{t+\Delta}, z), V^i(a_{t+\Delta}, z)\big\}\Big]\Big\}. \end{aligned} \tag{8}$$

We subtract $(1-\rho\Delta)V^i(a_t, z)$ from both sides of (8). On the left-hand side we obtain:

$$V^i(a,z) - (1-\rho\Delta)V^i(a,z) = \rho\Delta \cdot V^i(a,z). \tag{9}$$

On the right-hand side, the main term involving $V^i(a_{t+\Delta}, z)$ can be reorganized as:

$$\begin{aligned} &(1-\rho\Delta)\big[1-(\gamma^d+\gamma^z+\gamma^f)\Delta\big]V^i(a_{t+\Delta}, z) - (1-\rho\Delta)V^i(a,z) \\ &= (1-\rho\Delta)V^i(a_{t+\Delta}, z) - (1-\rho\Delta)(\gamma^d+\gamma^z+\gamma^f)\Delta V^i(a_{t+\Delta}, z) - (1-\rho\Delta)V^i(a,z) \\ &= (1-\rho\Delta)\Big[V^i(a_{t+\Delta}, z) - V^i(a,z)\Big] - (1-\rho\Delta)(\gamma^d+\gamma^z+\gamma^f)\Delta \cdot V^i(a_{t+\Delta}, z). \end{aligned} \tag{10}$$

Combining (9) and (10), regrouping all terms, and dividing by Δ, we obtain:

$$\begin{aligned}
\rho V^i(a_t, z) = \max_c \Bigg\{ & u(c) + (1-\rho\Delta)\frac{V^i(a_{t+\Delta}, z) - V^i(a_t, z)}{\Delta} \\
& -(1-\rho\Delta)(\gamma^d + \gamma^z + \gamma^f) \cdot V^i(a_{t+\Delta}, z) \\
& + (1-\rho\Delta)\gamma^d \cdot V^u(a_{t+\Delta}, z) \\
& + (1-\rho\Delta)\gamma^z \int V^i(a_{t+\Delta}, z')\, d\Pr_z(z'|z) \\
& + (1-\rho\Delta)\gamma^f \cdot \max\left\{V^f(a_{t+\Delta}, z),\ V^i(a_{t+\Delta}, z)\right\} \Bigg\} + O(\Delta). \qquad (11)
\end{aligned}$$

Note that the orange term combines with the Poisson inflow terms. For the job destruction terms:

$$\begin{aligned}
& -(1-\rho\Delta)\gamma^d V^i(a_{t+\Delta}, z) + (1-\rho\Delta)\gamma^d V^u(a_{t+\Delta}, z) \\
& = (1-\rho\Delta)\gamma^d \Big[V^u(a_{t+\Delta}, z) - V^i(a_{t+\Delta}, z)\Big]. \qquad (12)
\end{aligned}$$

Similarly for the productivity term:

$$\begin{aligned}
& -(1-\rho\Delta)\gamma^z V^i(a_{t+\Delta}, z) + (1-\rho\Delta)\gamma^z \int V^i(a_{t+\Delta}, z')\, d\Pr_z(z'|z) \\
& = (1-\rho\Delta)\gamma^z \int \Big[V^i(a_{t+\Delta}, z') - V^i(a_{t+\Delta}, z)\Big] d\Pr_z(z'|z). \qquad (13)
\end{aligned}$$

And for the formal offer term:

$$\begin{aligned}
& -(1-\rho\Delta)\gamma^f V^i(a_{t+\Delta}, z) + (1-\rho\Delta)\gamma^f \max\left\{V^f(a_{t+\Delta}, z),\ V^i(a_{t+\Delta}, z)\right\} \\
& = (1-\rho\Delta)\gamma^f \max\Big\{V^f(a_{t+\Delta}, z) - V^i(a_{t+\Delta}, z),\ 0\Big\}. \qquad (14)
\end{aligned}$$

Therefore, after combining:

$$\begin{aligned}
\rho V^i(a_t, z) = \max_c \Bigg\{ & u(c) + (1-\rho\Delta)\frac{V^i(a_{t+\Delta}, z) - V^i(a_t, z)}{\Delta} \\
& + (1-\rho\Delta)\gamma^d \Big[V^u(a_{t+\Delta}, z) - V^i(a_{t+\Delta}, z)\Big] \\
& + (1-\rho\Delta)\gamma^z \int \Big[V^i(a_{t+\Delta}, z') - V^i(a_{t+\Delta}, z)\Big] d\Pr_z(z'|z) \\
& + (1-\rho\Delta)\gamma^f \max\Big\{V^f(a_{t+\Delta}, z) - V^i(a_{t+\Delta}, z),\ 0\Big\} \Bigg\} + O(\Delta). \qquad (15)
\end{aligned}$$

Step 4: *Taking the Limit:* $\Delta \to 0$.

As $\Delta \to 0$, all $a_{t+\Delta} \to a_t = a$ and $(1-\rho\Delta) \to 1$. The key term we must evaluate is:

$$\lim_{\Delta \to 0} \frac{V^i(a_{t+\Delta}, z) - V^i(a_t, z)}{\Delta}. \tag{16}$$

We use the law of motion (5), which implies $a_{t+\Delta} = a_t + \Delta \dot{a}$:

$$\begin{aligned}
&\lim_{\Delta \to 0} \frac{V^i(a_{t+\Delta}, z) - V^i(a_t, z)}{\Delta} \\
&= \lim_{\Delta \to 0} \frac{V^i\big(a_t + \Delta(wv(z) + R_j a_t - c_t + b^i),\ z\big) - V^i(a_t,\ z)}{\Delta} \\
&= \lim_{\Delta \to 0} \frac{V^i\big(a_t + \Delta(wv(z) + r_j a_t - c_t + b^i),\ z\big) - V^i(a_t,\ z)}{\Delta(wv(z) + r_j a_t - c_t + b^i)} \frac{\Delta(wv(z) + r_j a_t - c_t + b^i)}{\Delta} \\
&= \lim_{\Delta \to 0} \frac{V^i\big(a_t + x,\ z\big) - V^i(a_t,\ z)}{x} \frac{(wv(z) + r_j a_t - c_t + b^i)}{1} \\
&= V^i_a(a_t, z) \cdot (wv(z) + r a_t - c_t + b^i).
\end{aligned} \tag{17}$$

Defining $x \equiv \Delta \dot{a}$, as $\Delta \to 0$ we have $x \to 0$, and the first factor is by definition the partial derivative of V^i with respect to its first argument:

$$\lim_{x \to 0} \frac{V^i(a + x, z) - V^i(a, z)}{x} = V^i_a(a, z).$$

Therefore:

$$\lim_{\Delta \to 0} \frac{V^i(a_{t+\Delta}, z) - V^i(a, z)}{\Delta} = V^i_a(a, z) \cdot \dot{a}. \tag{18}$$

Step 5: *The HJB Equation in Continuous Time.*

Substituting (18) into (15) and taking $\Delta \to 0$, we obtain the Hamilton-Jacobi-Bellman equation in continuous time:

$$\begin{aligned}
\rho V^i(a, z) = \max_c \Big\{ & u(c) + V^i_a(a, z)\dot{a} + \gamma^d \Big[V^u(a, z) - V^i(a, z)\Big] \\
& + \gamma^z \int \Big[V^i(a, z') - V^i(a, z)\Big] d \Pr_z(z'|z) \\
& + \gamma^f \max\Big\{V^f(a, z) - V^i(a, z),\ 0\Big\} \Big\}
\end{aligned} \tag{19}$$

subject to:

$$\begin{aligned}
&\dot{a} = w \cdot v(z) + ra - c + b^i, \\
&a \geq \underline{a}.
\end{aligned}$$

Additionally, the voluntary participation constraint establishes that the agent remains an informal worker only if doing so is at least as good as the alternatives to which the agent can freely switch:

$$V^i(a, z) \geq \max\Big\{V^u(a, z),\ V^{ie}(a, z),\ V^{fe}(a, z)\Big\}. \tag{20}$$

Step 6: *Interpretation of Each Term.*

Equation (19) admits the following interpretation. The left-hand side, $\rho V^i(a,z)$, is the "opportunity cost" of being in the current state—the return one would obtain by investing the value V^i at the discount rate ρ. The right-hand side collects all flows and risks:

1. $u(c)$: the instantaneous flow utility from consumption.
2. $V_a^i(a,z) \cdot \dot{a}$: the capital gain from asset accumulation (or decumulation). It is the product of the "marginal value of wealth" and the saving rate.
3. $\gamma^d[V^u(a,z) - V^i(a,z)]$: the expected effect of job destruction. At rate γ^d the agent transitions to unemployment, and the change in value is $V^u - V^i$ (typically negative).
4. $\gamma^z \int [V^i(a,z') - V^i(a,z)]\, d\,\mathrm{Pr}_z(z'|z)$: the expected effect of productivity shocks. The integral averages the change in value over all possible realizations z' of the new productivity.
5. $\gamma^f \max\{V^f(a,z) - V^i(a,z), 0\}$: the "option value" of the formal job offer. At rate γ^f an offer arrives, and the agent accepts it only if $V^f > V^i$; otherwise, the term equals zero.

Step 7: *Remark: Poisson vs. Diffusion*

If instead of a Poisson process with discrete jumps, productivity z followed a diffusion process of the Ornstein-Uhlenbeck type:

$$dz = -\theta(z - \bar{z})\, dt + \sigma\, dW_t, \tag{21}$$

where W_t is a standard Brownian motion, then Itô's lemma would imply that the HJB takes the form:

$$\begin{aligned} \rho V^i(a,z) = \max_c \Big\{ & u(c) + V_a^i(a,z) \cdot \dot{a} + \gamma^d \Big[V^u(a,z) - V^i(a,z) \Big] \\ & + \underbrace{-\theta(z-\bar{z}) \cdot V_z^i(a,z)}_{\text{drift}} + \underbrace{\tfrac{1}{2}\sigma^2 \cdot V_{zz}^i(a,z)}_{\text{volatility}} \\ & + \gamma^f \max \Big\{ V^f(a,z) - V^i(a,z),\ 0 \Big\} \Big\}. \end{aligned} \tag{22}$$

The fundamental difference is the appearance of two new terms: V_z^i (the effect of the drift or tendency of z) and $\frac{1}{2}\sigma^2 V_{zz}^i$ (the effect of volatility, which captures how the concavity of the value function with respect to z interacts with uncertainty—essentially Jensen's inequality).

In practice, the Poisson approach with a discrete grid replaces the integral $\int [V^i(a,z') - V^i(a,z)]\, d\,\mathrm{Pr}_z(z'|z)$ with a finite sum over n_z grid points, transforming the PDE into a system of coupled ODEs (one for each point on the z grid), which is computationally more efficient. The Rouwenhorst (1995) method, used by Herreño & Ocampo (2023), approximates an AR(1) process in $\log(z)$—the discrete-time counterpart of (21)—by a Markov chain with n_z states, which in continuous time is represented as a Poisson process with multiple states.